

Beyond Correctness: A Controlled Benchmark of AI Models on Advanced University Mathematics and Physics

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Abstract. Open-ended university problems expose whether a model can construct and verify a solution rather than merely recognize an answer. The objective is to compare solution correctness and completeness together with the resources required to obtain them. A controlled, closed-resource benchmark evaluates 14 model–mode configurations on 60 authentic examination problems in Mathematical Analysis III and General Physics II. Sixty item-specific rubrics contain 1,600 atomic criteria; answers are assessed anonymously in single-response contexts, centrally validated, and every case of substantive doubt is escalated to versioned human adjudication. The six leading configurations occupy a narrow 96.71–98.72% band, and paired bootstrap intervals do not separate the first system from its two nearest competitors. In a same-weights DeepSeek V4 Flash contrast, high reasoning adds 10.93 percentage points (95% CI 7.97–14.10), while increasing token use by 214.7%, cost by 224.4%, and median latency 5.13-fold. Across the normalized matched panel, model-level analytical-token volume, estimated API cost, and median latency vary by factors of 6.91, 234.61, and 12.46. These results show that, once correctness approaches saturation, resource demand, robustness across domains, and quality–resource frontiers become central outcomes.

Keywords: large language models; mathematical reasoning; physics problem solving; blind rubric evaluation; efficiency; latency.

1 Introduction

Scientific problem solving is an unusually demanding test of language models. An acceptable response must identify the mathematical or physical structure, select a defensible method, execute a multi-step derivation, respect domains or units, and communicate a complete conclusion. Exact-match answers collapse these requirements into one bit: they miss invalid derivations that happen to end correctly and substantial correct work followed by a local error.

The same limitation now affects model ranking. When several systems solve most items correctly, small differences in mean score may be weaker than their sampling uncertainty, while latency, hidden reasoning, and price can differ by orders of magnitude. A benchmark intended for model selection must therefore measure not only whether a solution is correct but also how it is correct and what is consumed to produce it.

This study addresses three questions. *RQ1* asks how accurately and completely 14 current configurations solve the same 60 advanced examination problems, including the residual criterion-level structure behind their scores. *RQ2* asks whether performance is stable between Mathematical Analysis III and General Physics II. *RQ3* asks how quality relates to token volume, dated estimated API cost, and end-to-end response time. One same-weights thinking/no-thinking pair additionally provides a focused config-

uration contrast.

The contribution is a traceable measurement architecture rather than a larger question bank. Authentic examination material is combined with a direct-API, one-item, no-tool execution condition; frozen item-specific atomic rubrics; blind single-answer assessment; permanent response-to-identity mappings; central validation; and human authority resolution before unblinding. The result is a balanced 14-by-60 panel in which correctness and resource measurements refer to exactly the same model–item cells.

2 Related Work and Positioning

Scientific benchmarks span broad college-level problem solving, expert-written graduate questions, undergraduate physics coverage, open-answer validation, and step-level reasoning. SciBench and GPQA established complementary breadth and expert difficulty; UGPhysics broadened undergraduate physics and hybrid grading; PHYSICS and PhysReason emphasized open derivations and intermediate reasoning (Wang et al., 2024; Rein et al., 2024; Xu et al., 2025; Feng et al., 2025; Zhang et al., 2025). More recent benchmarks emphasize original problem construction or multimodal decompositions of variables, processes, and solutions (Qiu et al., 2025; Dai et al., 2025). The present study is smaller by design: its unit is an authentic examination exercise evaluated

in depth against a frozen, inspectable scoring contract.

Benchmark governance is equally important. HELM motivates standardized conditions and multi-dimensional reporting; BetterBench emphasizes life-cycle documentation and reproducibility; contamination studies and LiveBench caution against treating static scores as timeless evidence (Liang et al., 2023; Reuel et al., 2024; Deng et al., 2024; White et al., 2025). Here, exact prompts, configurations, raw outputs, judgments, adjudications, exclusions, and dated resource prices are part of the measured artifact chain.

Model-based evaluation makes criterion-level grading scalable, but it can introduce position, verbosity, self-preference, and self-recognition biases (Zheng et al., 2023; Hashemi et al., 2024; Panickssery et al., 2024; Verga et al., 2024). Model judgments are therefore treated as provisional measurements rather than final authority. Solver identities and competing responses are hidden, rubrics are fixed before scoring, every judgment must cite evidence and pass deterministic validation, and substantive uncertainty is escalated to human adjudication. Reliance on a single judge family nevertheless remains a limitation.

Table 1 positions the present protocol through documented design choices rather than binary feature claims or an implied ranking of benchmark quality.

No individual element in the final row is claimed as unique. The contribution is their integration within one auditable authority chain: authentic dated examinations, frozen atomic rubrics, blind judgments, mandatory human resolution of substantive doubt, and resource measurements all refer to the same cells.

3 Benchmark Design

3.1 Corpus and task condition

The corpus contains 20 examination papers from the 2025 and 2026 sessions of two advanced courses in a physics degree. Each paper contributes exactly three open-ended exercises. The balanced population contains 33 Analysis III items and 27 Physics II items. Analysis problems cover differential manifolds, geometry, and differential equations; Physics II covers electromagnetism, circuits, waves, and geometrical optics. Every item is linked to its official solution and an approved operational rubric.

Original examination PDFs are retained as provenance records, while statements and official solutions are transcribed to UTF-8 without silent cor-

rection. Any disagreement among the scan, transcription, official solution, and rubric enters human review. Essential figures were replaced by one authorized textual description used unchanged for generation, rubric construction, and judgment. The resulting track measures reasoning from controlled text, not visual perception. Each solver received one exercise per call and a fixed response contract requiring assumptions, method, verifiable development, separated final results, and relevant checks. Web access, retrieval, code execution, calculators, and all other tools were prohibited both in the prompt and in the API configuration.

Table 2. Certified analysis population. Responses include repeated calls; model-item cells are the macro-averaged unit used in comparisons.

Component	Count
Examination papers / exercises	20 / 60
Analysis III / Physics II items	33 / 27
Model-mode configurations	14
Active responses, including replications	924
Balanced model-item cells	840
Frozen rubrics / rubric subproblems	60 / 214
Atomic scoring criteria	1,600
Open human-review cases at snapshot	0

3.2 Configurations and direct-API execution

All generation requests were sent by the benchmark harness directly to the first-party API of the selected provider. No aggregator, including OpenRouter, was used. A versioned configuration fixed the model identifier, reasoning mode, item set, prompt, repetition policy, concurrency, timeout, retry policy, output ceiling, and dated price table. The effective prompt, normalized answer, raw provider object, and technical metadata were persisted separately. Thus, what the model saw, what the judge saw, and what the provider returned remain independently reconstructible.

Table 3 reports the exact identities in the balanced panel. Active response counts exceed 60 for configurations that participated in early campaign waves with three calls on selected 2025 items to observe within-cell stochastic repeatability. Later completion waves used one call per missing cell to extend matched coverage within the available budget. This wave-level design choice was independent of answer quality, and replications are never treated as new items.

Table 1. Protocol-level positioning against representative scientific-reasoning benchmarks cited above. Entries summarize documented task and evaluation designs; they do not rank overall benchmark quality.

Benchmark	Task and source	Response and scoring design	Human and operational scope
SciBench (Wang et al., 2024)	College mathematics, chemistry, and physics; textbooks plus a smaller course-exam subset	Open solutions with primarily final-answer assessment	Broad collegiate coverage and manual error analysis
GPQA (Rein et al., 2024)	Expert-authored graduate biology, chemistry, and physics questions	Four-option multiple choice with exact-choice accuracy	Expert item validation and revision
UGPhysics (Xu et al., 2025)	Undergraduate physics exercise books covering multiple subjects	Open solutions and seven answer types; rule- and model-based grading	Human validation of evaluator samples and token analyses
PHYSICS (Feng et al., 2025)	Public physics qualifying examinations	Open solutions; regular-expression or symbolic checks with model fallback	Expert annotation and cross-checking
PhysReason (Zhang et al., 2025)	Entrance, practice, and competition physics problems	Answer- and step-level scoring of open reasoning	Human validation of evaluator outputs
This work	60 items from 20 dated university course examinations	Open derivations; frozen atomic rubrics and blind single-answer judging	Case-level human adjudication and paired token, cost, and latency measurements

Table 3. Exact model–mode configurations. “Active responses” includes replications before within-cell macro-averaging. API identifiers and campaign-level modes are reported as configured.

Provider	Display name	Exact API identifier	Configured mode	Active responses
OpenAI	GPT-5.6 Sol	gpt-5.6-sol	reasoning, xhigh	78
OpenAI	o3	o3-2025-04-16	reasoning, high	60
OpenAI	GPT-4.1	gpt-4.1-2025-04-14	no native reasoning	60
Anthropic	Claude Fable 5	claude-fable-5	effort, high	78
Anthropic	Claude Sonnet 4.5	claude-sonnet-4-5-20250929	provider default	60
Anthropic	Claude Haiku 4.5	claude-haiku-4-5-20251001	no thinking parameter requested	60
Google	Gemini 3.1 Pro Preview	gemini-3.1-pro-preview	native reasoning	78
Google	Gemini 3.5 Flash	gemini-3.5-flash	thinking, high	90
Google	Gemma 4 31B IT	gemma-4-31b-it	thinking disabled	60
DeepSeek	V4 Flash	deepseek-v4-flash	thinking enabled, reasoning high	60
DeepSeek	V4 Flash (no thinking)	deepseek-v4-flash	thinking disabled	60
DeepSeek	V4 Pro	deepseek-v4-pro	thinking enabled, reasoning high	60
xAI	Grok 4.5	grok-4.5	reasoning, high	60
xAI	Grok 4.3	grok-4.3	reasoning, high	60

3.3 Run provenance and resource boundaries

Metadata record timestamps, attempts, finish reason, latency, provider token fields, cost provenance, and prompt and response hashes. Resume and retry never overwrite a completed historical artifact. The repository-wide index contains 977 run records.¹ The balanced analysis therefore retains 924 successful responses belonging to the 14 configurations in table 3, then reduces them to 840 equally weighted model–item cells before between-model comparison.

Analytical tokens are input plus visible output

¹Six are failed unavailable-endpoint calls, and 47 are completed Gemini 3.1 Flash-Lite runs excluded because that configuration does not cover the complete 60-item panel. Among the 924 retained responses, 798 model–item cells have one run and 42 have three: nine cells each for GPT-5.6 Sol, Claude Fable 5, and Gemini 3.1 Pro Preview, and 15 for Gemini 3.5 Flash. The 84 additional runs are averaged within their cells.

plus separately reported reasoning. For Claude Fable 5, the provider exposes a combined output/thinking quantity, which is retained as an un-separated class. Costs are provider-reported for 120 of the 840 cells and reconstructed from versioned campaign rates for 720. Resource totals used for between-model comparison are computed after within-cell macro-averaging and therefore describe an equal-weight 840-cell panel, not billed campaign expenditure. Before averaging, the 924 retained successful responses correspond to 10,054,390 analytical tokens and an estimated \$105.10 under the same hybrid accounting; this still excludes unobserved failed attempts and calls outside the matched panel. At least one request required retry in 22 cells. Token and cost fields record the final successful generation only, because consumption from failed attempts is not available. Latency is measured from

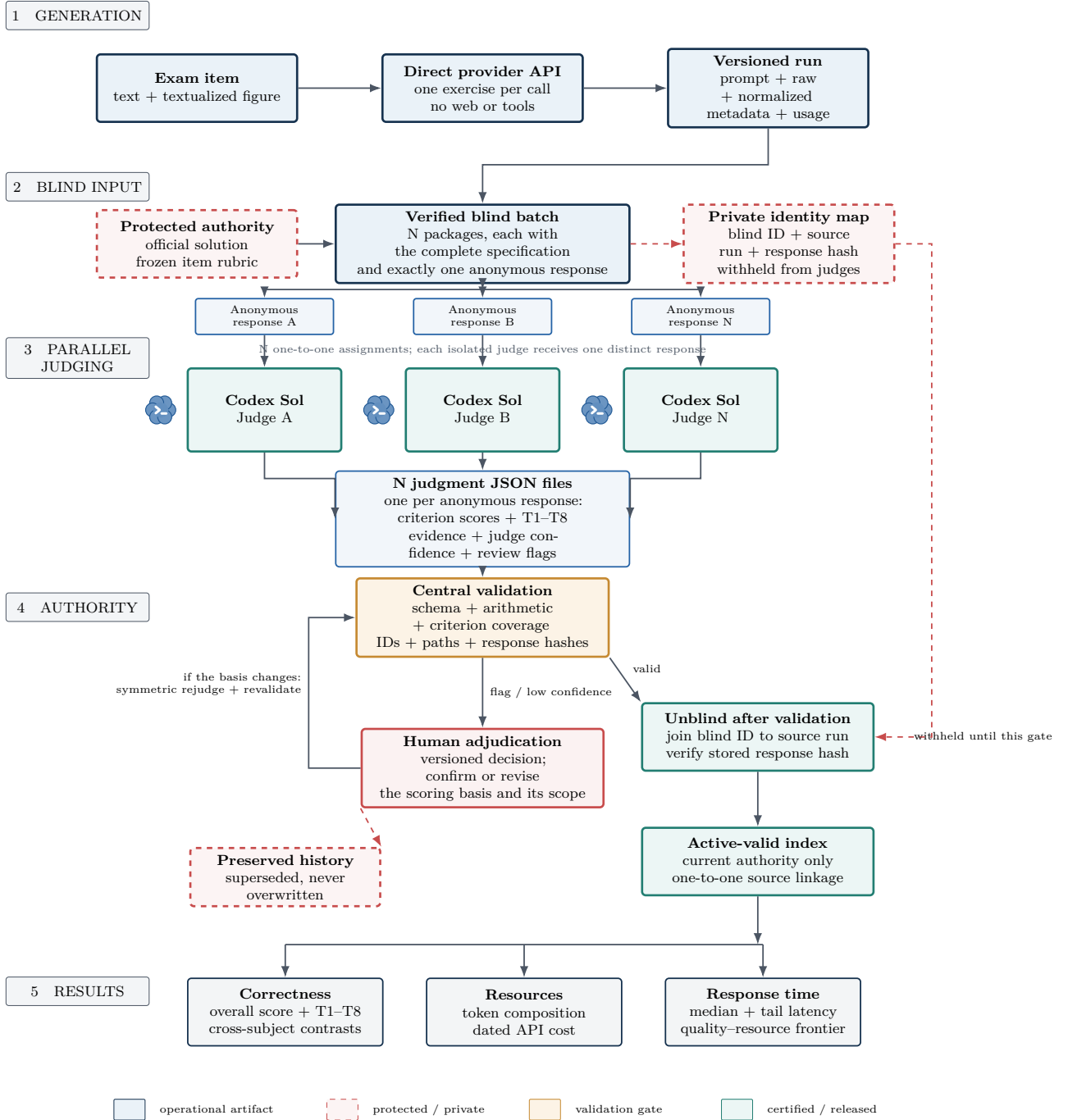


Figure 1. Benchmark architecture and authority chain. The judging lanes denote distinct anonymous responses: a batch of N responses produces N one-to-one assignments, each handled in an isolated context, rather than three judgments of the same answer. Generation is separated from evaluation, central validation gates unblinding, and human review resolves substantive doubt. Only a change to the common scoring basis triggers symmetric rejudging of all affected answers. The three result branches correspond to the analyses in section 5.

the first dispatch to the successful return, so it includes provider processing, backoff, and retry delays but not time spent waiting for a local concurrency slot.

4 Blind Evaluation and Analysis

4.1 Frozen atomic rubrics

Rubrics were constructed from the item, official solution, and authorized figure description before model answers were examined. The 60 rubrics contain 214 weighted subproblems and 1,600 atomic criteria: 837 criteria for the 33 Analysis III items and 763 for the 27 Physics II items. Each criterion specifies full-

credit evidence, four partial-credit anchors, zero conditions, accepted equivalences, alternative methods, and propagated-error policy. Physics criteria additionally encode tolerances, units, directions, phases, and peak/RMS conventions. Style and verbosity receive no points.

Criteria map to the eight dimensions in table 4. The final item score is the percentage of points applicable to that exercise. T8 is applicable to 14 Analysis III items only; it is not defined for the remaining 19 Analysis III items or any Physics II item. This conditional denominator is preserved in all profiles rather than imputing zero.

4.2 Blind judging, validation, and human authority

Each first-pass judge received one anonymous response together with the full subject specification, item, official solution, and frozen rubric. It did not receive solver identity, competing answers, prior scores, or the private map. For a batch of N responses, the system created N isolated Codex Sol assignments: each assignment scored one distinct response, cited evidence for every applicable criterion, calculated T1–T8, and emitted one structured judgment. Before dispatch, every blind ID was bound permanently to one source run and response hash; bijective map and manifest checks preceded the first judgment.

Central validation then checked schema, criterion coverage, arithmetic, dimension sums, identifiers, unique paths, allowed values, and frozen-input hashes. Only validated judgments could be unblinded. Blind single-answer assessment prevents position effects and direct cross-model comparison at the judging stage, but a single model-family judge can still introduce systematic bias; this remains a limitation rather than being assumed away.

Every case of substantive doubt was reviewed by the human benchmark lead before unblinding; no flagged case was resolved by the model judge alone. The reviewer can confirm the frozen basis, recognize an equivalent or stronger method, correct a reference or rubric, or identify a judge error. Decisions are versioned and historical files remain preserved. Crucially, confirmation of the existing basis changes only the flagged judgment, whereas a material change to that basis initiates symmetric blind reassessment of every affected answer. At the analysis snapshot, all planned judgments validated and unblinded, with no unresolved human-review flag.²

²In the 282-response completion wave—the final batch used to fill missing cells in the 14-by-60 panel—21 responses were blindly rejudged on previously approved adjudicated bases. Five received score-preserving overlays, meaning ver-

4.3 Aggregation and uncertainty

The exercise is the sampling unit. For each configuration, repeated responses are first averaged within their model–item cell, producing one score for each of the same 60 exercises. The reported model mean is the arithmetic mean of those 60 cell scores, so a triplicated cell has the same weight as a cell with one response and every configuration remains directly paired by item.

Uncertainty is estimated with 20,000 nonparametric bootstrap iterations using seed 20260810. In each iteration, 60 exercise indices are sampled with replacement from the common item set. For every configuration, the mean is recomputed from the cell scores at those sampled indices; the 2.5th and 97.5th percentiles form the marginal 95% interval. For a paired model contrast, the same resampled indices are applied to both configurations before their means are subtracted, preserving item-level pairing in every iteration. These intervals quantify sensitivity to sampling exercises from this benchmark; they do not include uncertainty from future provider changes or from replacing the judge model.

Replications and bootstrap resampling address different uncertainties: the 42 triplicated cells observe generation variability for selected model–item pairs, whereas the bootstrap resamples the 60 exercise identities to estimate item-sampling uncertainty without imputing run variability to single-run cells.

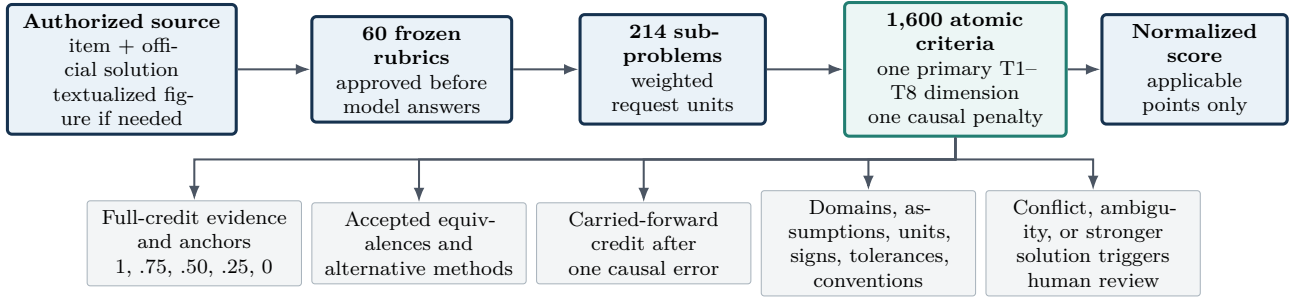
Resource totals and subject means are calculated on the same macro-averaged cells. Pareto fronts maximize score while minimizing one resource at a time; no composite efficiency index or subjective exchange rate is imposed. Correlations use the 14 model aggregates and are reported as descriptive associations only.

5 Results

5.1 A near-ceiling leading group

Table 5 is the numerical backbone of the results: every row summarizes the same 60 items and joins quality to resource demand. The first six configurations occupy only 2.01 percentage points, from 96.71% to 98.72%. Their marginal intervals overlap substantially. More directly, the paired difference between GPT-5.6 Sol and Claude Fable 5 is 0.72 pp (95% CI -0.56 to 1.95), and the difference between GPT-5.6 Sol and DeepSeek V4 Flash is 1.48 pp (95% CI -0.23 to 3.73). The observed order should therefore not be read as fourteen precisely

sioned annotations that incorporated an approved decision without changing the numerical score; three run-validity classifications were overruled, also without score changes.



No style or verbosity criterion; frozen authorities cannot be silently reinterpreted by a judge.

Figure 2. Anatomy of the scoring contract. A criterion is answer-independent, belongs to one primary technical dimension, and can penalize one causal error only once. Judges may flag a reference or rubric conflict, but they cannot silently reinterpret a frozen authority.

Table 4. Technical dimensions and nominal budgets. The same high-level dimensions govern both subjects; criterion evidence remains item-specific.

Dim.	Pts	Construct	Dim.	Pts	Construct	Dim.	Pts	Construct	Dim.	Pts	Construct
T1	20	Understanding / model	T2	15	Method selection	T3	20	Development	T4	15	Results / coverage
T5	10	Conditions / conventions	T6	8	Coherence / plausibility	T7	7	Theorems / laws	T8	5	Independent verification

Table 5. Main matched-panel results. Score intervals are 20,000-resample item-bootstrap intervals. Tokens and estimated cost are normalized panel totals for 60 macro-averaged cells; time is the median model–item response time. Shading marks the six-model near-ceiling group.

Rank	Configuration	Overall score [95% CI]	Analysis III	Physics II	Tokens (M)	Est. cost (USD)	Median (s)
1	GPT-5.6 Sol	98.72 [97.04, 99.86]	99.86	97.33	0.959	27.8053	307.0
2	Claude Fable 5	97.99 [96.15, 99.38]	98.44	97.45	0.507	22.9185	83.5
3	DeepSeek V4 Flash	97.24 [94.66, 99.16]	97.10	97.41	1.396	0.3845	188.0
4	Gemini 3.1 Pro Preview	97.05 [95.11, 98.60]	96.63	97.56	0.690	7.8895	114.7
5	Gemini 3.5 Flash	96.95 [95.10, 98.51]	97.29	96.53	0.976	8.4965	61.2
6	DeepSeek V4 Pro	96.71 [94.76, 98.31]	96.57	96.89	1.066	0.9102	202.4
7	Grok 4.5	94.61 [91.66, 97.08]	95.59	93.42	0.576	3.2504	113.8
8	OpenAI o3	92.75 [89.74, 95.33]	90.49	95.52	0.644	4.9216	70.2
9	DeepSeek V4 Flash (no thinking)	86.31 [82.73, 89.68]	84.73	88.24	0.443	0.1185	36.6
10	Gemma 4 31B IT	85.58 [81.52, 89.34]	82.80	88.98	0.209	0.0000 (cfg.)	74.8
11	Grok 4.3	85.24 [79.77, 90.11]	80.40	91.16	0.656	1.5744	73.7
12	OpenAI GPT-4.1	71.58 [66.46, 76.53]	64.02	80.82	0.311	2.2599	39.1
13	Claude Sonnet 4.5	70.91 [65.31, 76.41]	60.71	83.37	0.202	2.4706	37.4
14	Claude Haiku 4.5	64.77 [59.57, 69.80]	57.59	73.54	0.279	1.2084	24.6

cfg. marks a campaign-configured zero, not a general market price.

separated performance levels. “Top-six” is an empirical near-ceiling label based on close estimates and overlapping intervals, not a formal equivalence class; no equivalence margin was prespecified.

Across all 840 cells, 41.67% are perfect and 91.19% score at least 60%; the macro-mean is 88.31%. These aggregates describe the selected panel, not an expected score for an arbitrary model or course.

Figure 5 shows that saturation is not uniform. Exercise means span 50.84–99.68%; 30 of 60 remain below 90% and nine below 80%, seven of them in Analysis III. The corpus therefore combines a broad high-scoring region with a smaller, substantively dis-

criminating tail.

5.2 Residual structure and cross-domain robustness

The dimension panel in figure 3 identifies what the scalar score hides. Results and coverage (T4) is the weakest fully comparable dimension in both subjects. The common residual weakness is therefore not merely selecting a method; it is carrying the response through every requested result. T8 is shown separately because its restricted 14-item denominator is not comparable with the populations used for T1–T7.

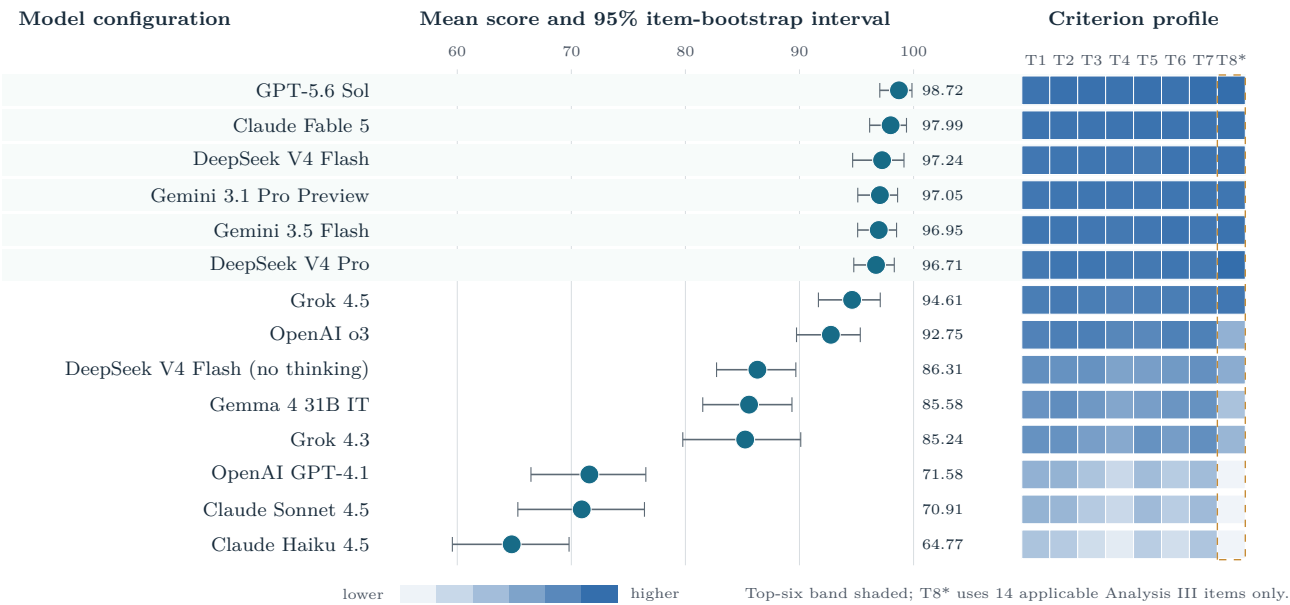


Figure 3. Overall performance and criterion profile on the balanced 14-by-60 panel. Left: mean score with marginal 95% item-bootstrap interval; values are printed at the right edge. Right: sequentially shaded T1–T8 means, with the constructs defined in table 4. The dashed T8 column is not directly comparable with T1–T7 because verification is applicable on only 14 Analysis III items. Overlapping intervals at the top motivate treating the first six configurations as a near-ceiling group rather than fourteen sharply resolved ranks.

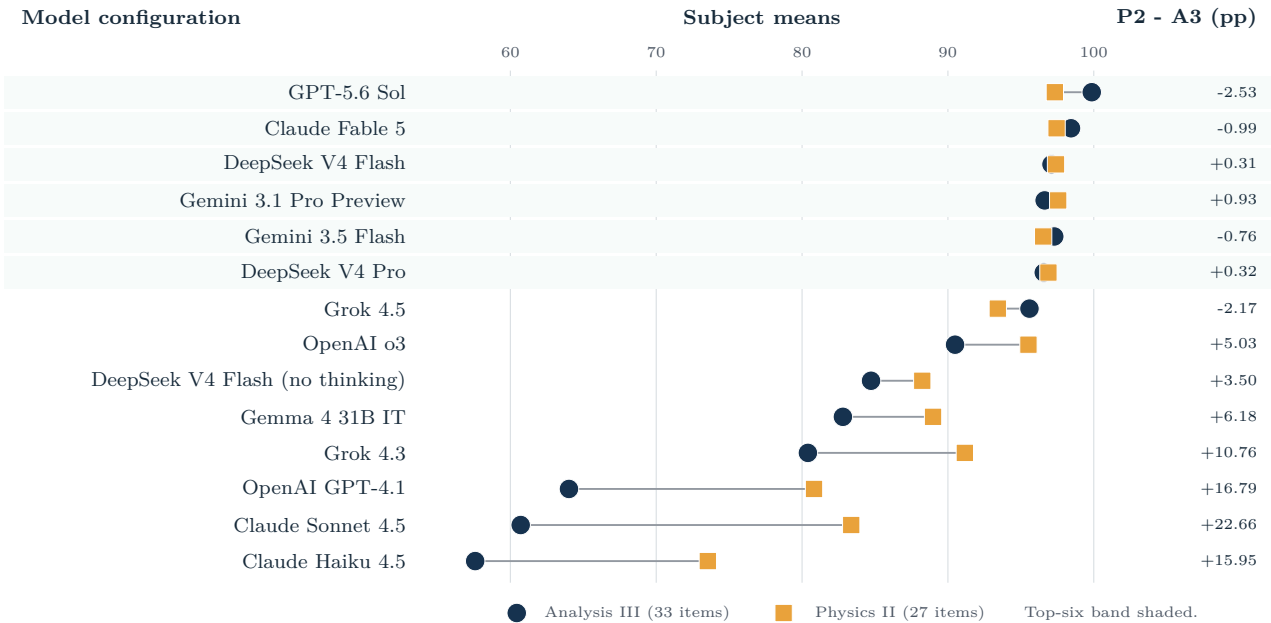


Figure 4. Cross-domain score heterogeneity. Subject means and their paired difference are reported on a common scale; because the two subjects contain different exercises, the gaps are descriptive domain contrasts rather than causal estimates of subject difficulty.

The current judgment ledger records 1,211 atomic errors across 508 retained responses, predominantly at major severity. Because these counts are response-level, replicated cells receive more weight and the ledger is not used to rank models. Raw category labels are also not yet a controlled taxonomy. Severity and the frozen T1–T8 mapping are the defensible aggregate views; ad hoc frequency claims from synonymous category strings are avoided.

The paired view in figure 4 shows that the leading configurations remain stable across subjects,

whereas several lower-ranked systems perform substantially better in Physics II than in Analysis III. This domain sensitivity would be obscured by separate subject rankings.

5.3 Reasoning is valuable, but not free

DeepSeek V4 Flash is the cleanest configuration contrast because the API identifier, weights, prompt, and 60 items are held fixed while thinking is toggled. High reasoning raises the score from 86.31%

to 97.24%, a paired gain of 10.93 pp (95% CI 7.97–14.10), while substantially reducing the share of responses containing a recorded atomic error.

The resource changes in figure 6 show that the gain requires more than three times the token volume and estimated cost and more than five times the median response time. Visible output nevertheless decreases, indicating that the additional volume is internal reasoning rather than a longer displayed derivation. The conclusion is configuration-specific: high reasoning is decisive for this model on these tasks, but it shifts every measured resource coordinate.

5.4 Quality–resource frontiers

After within-cell averaging, the matched comparison panel represents 8,915,801 analytical tokens and an estimated normalized cost of \$84.21. These are equal-weight comparison totals, not a provider bill: before averaging, the 924 retained successful responses correspond to 10,054,390 analytical tokens and an estimated \$105.10. Input accounts for 6.80% of normalized volume, visible output 28.41%, separately reported reasoning 59.79%, and Fable’s inseparable output/thinking field 5.00%; model totals span 6.91-fold. Non-zero normalized model cost spans \$0.1185–\$27.8053 (234.61-fold), and GPT-5.6 Sol plus Fable account for 60.24% of the normalized comparison cost. Token volume and price are only weakly associated (Spearman $\rho = 0.23$), reflecting provider-specific rate structures.

Median response time varies 12.46-fold; GPT-5.6 Sol is the slowest configuration by this measure, at 307.05 seconds. Across the 14 models, score is associated with log median latency (Pearson $r = 0.80$; Spearman $\rho = 0.78$), but this small-sample relationship is neither item-level nor causal. Gemini 3.5 Flash and Fable are operationally important exceptions, combining near-ceiling scores with markedly lower medians. Retry-heavy tails also make means less representative for several configurations.

The accuracy–cost front contains Gemma 4, both DeepSeek Flash modes, Fable, and GPT-5.6 Sol. The token front substitutes Sonnet 4.5 for thinking DeepSeek; the latency front contains Haiku, no-thinking DeepSeek, Gemini 3.5 Flash, Fable, and GPT-5.6 Sol. Front membership is campaign-specific, not a deployment recommendation.

6 Discussion

6.1 What a near-ceiling score does and does not mean

The central result is a plateau, not an isolated winner. The best point estimate is 98.72%, yet the paired uncertainty around differences within the leading group is large enough that tenths of a point should not drive model selection. At the same time, the plateau is not evidence that the benchmark has become uninformative. Criterion profiles identify completeness as the most persistent fully comparable weakness, cross-domain gaps distinguish robust systems from subject-sensitive ones, and the DeepSeek contrast shows that a reasoning setting can move a configuration from the middle of the table into the near-ceiling band.

The pace of progress is visible in the status reversal of GPT-4.1. At its April 2025 launch, it was presented as OpenAI’s leading coding model and as outperforming GPT-4o across the board (OpenAI, 2025); in this benchmark, the fixed no-reasoning GPT-4.1 snapshot scores 71.58%. Only sixteen months later, the six leading configurations reach 96.71–98.72% on the same 60 exercises, an absolute advantage of 25.13–27.14 percentage points. A system at the frontier of the preceding model generation therefore leaves more than one quarter of the rubric score unrealized on problems that current leaders solve at near-perfect mean accuracy. Because model families and inference modes differ, this is not a controlled longitudinal estimate; it is direct evidence, on a common item set, of the scale of capability progress over a short development interval.

Saturation changes the decision problem. Accuracy remains the admission criterion, but systems above a required threshold can then be compared on latency, token budget, price, or domain stability. The Pareto views preserve these trade-offs instead of hiding them in one arbitrary weighted index.

6.2 Why human intervention is part of the measurement

Open-ended mathematics does not always have one canonical path. A response may use an unlisted equivalence, carry a local error through otherwise valid work, or prove a result stronger than the course reference. Treating the frozen reference as infallible would turn these cases into systematic benchmark error; allowing the model judge to rewrite it silently would make the scoring contract unstable. The authority loop in figure 1 provides a third option: flag uncertainty, obtain a reasoned human decision, preserve the former version, and rejudge symmetrically

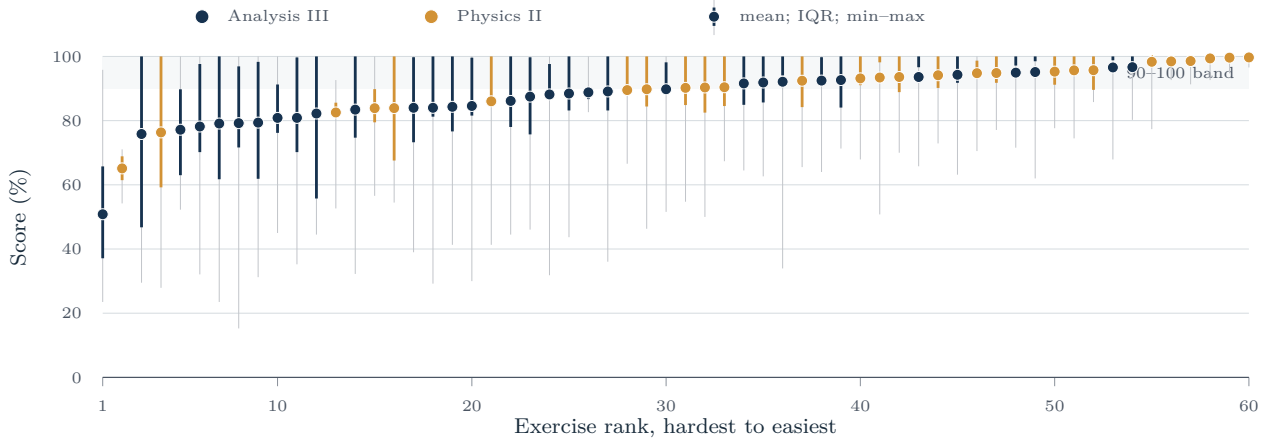


Figure 5. Difficulty profile of the 60 exercises. Items are ordered by their mean score across the 14 configurations. Points mark means, thick segments the interquartile range, and thin whiskers the full inter-model range; color denotes subject. The shaded 90–100% band makes the broad ceiling region visible while retaining the smaller tail of strongly discriminating exercises.

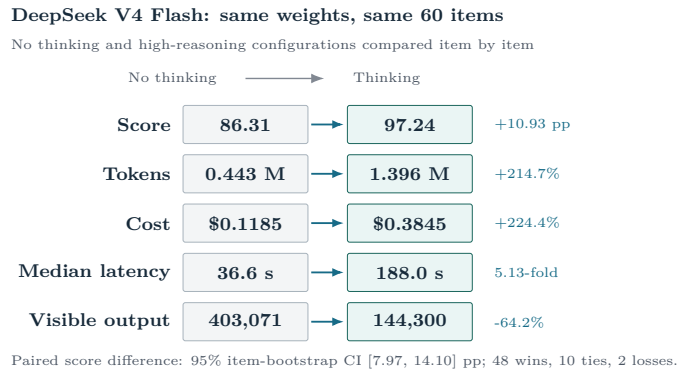
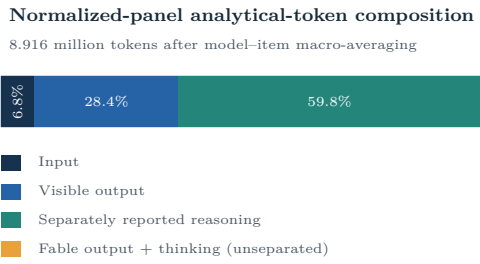


Figure 6. Token accounting and the DeepSeek thinking contrast. Left: aggregate token composition; Fable output and hidden thinking are inseparable in the available provider field. Right: paired same-weights comparison; the score interval resamples items, whereas resource deltas are configuration-specific summaries.

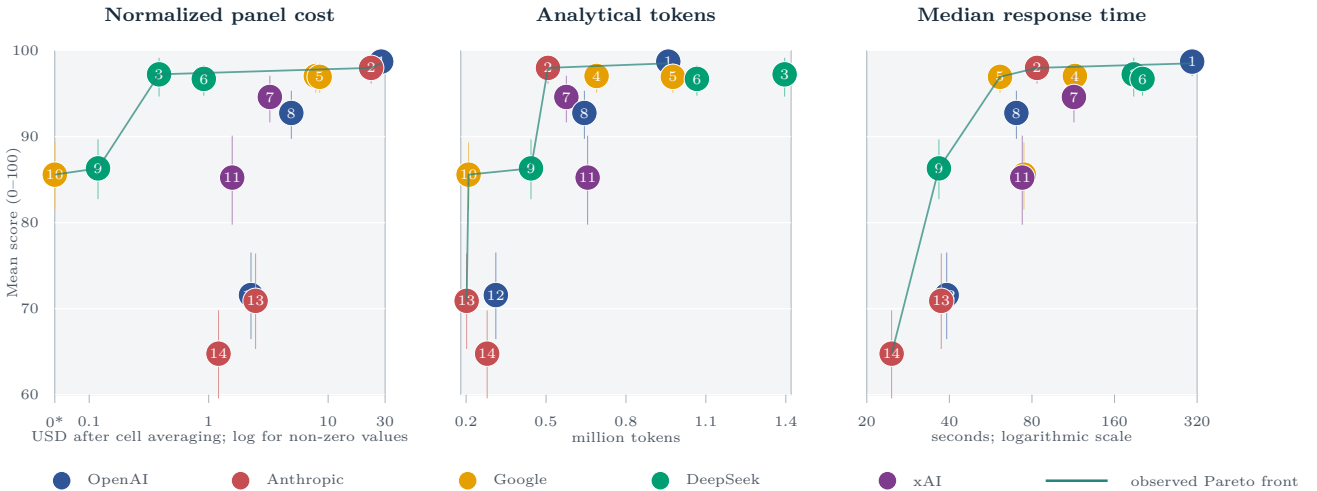


Figure 7. Accuracy–resource views for the same 14 configurations. Point numbers are ranks from table 5; vertical segments reproduce score intervals. Cyan lines mark the observed two-dimensional Pareto fronts, not fitted curves. The cost panel uses the normalized, cell-averaged estimates, treats Gemma’s campaign-specific zero as a separate point, and applies a logarithmic scale only to non-zero values. A system can be efficient in one panel and inefficient in another, so no composite “winner” is inferred.

only when the common basis changes.

This design does not remove human judgment. It makes human review mandatory whenever uncertainty concerns equivalence, reference authority,

rubric scope, or judge error, and makes the consequences inspectable. Zero open review cases at the snapshot means the release is resolved, not that human oversight was unnecessary.

6.3 Temporal evidence and possible training exposure

Release, knowledge, and training-data cutoffs are not interchangeable: the first establishes endpoint availability, while the other two retain the provider’s own scope and terminology. Month-only cutoffs are treated conservatively: an item counts as later only when its examination date follows the final day of that month.

The dated comparisons in table 6 show substantial temporal separation for many item–model pairs, especially for systems with 2024 or early-2025 meta-data. Later systems retain fewer post-cutoff items, but the table still documents nonzero separation where official dates are available.

This date-based comparison reduces the plausibility of straightforward pre-training exposure for those item–model comparisons, but it is not a contamination-free certificate. Provider labels are not standardized; later post-training, synthetic or private data, user-contributed material, and indirect exposure cannot be excluded. No analogous calculation is made for DeepSeek V4 or Grok 4.3 because an official model-specific cutoff was not located.

7 Limitations

Scope and saturation. The corpus contains 60 items from two courses at one institution. Authenticity supports ecological validity for this curriculum, but breadth is narrower than large synthetic or multi-institutional benchmarks. The leading group is close to the scale ceiling, and 60 items leave appreciable uncertainty for differences near one percentage point. Future waves should add newly authored, more discriminating exercises while preserving longitudinal anchors.

Text-only execution. Authorized descriptions replace essential figures, so the benchmark does not test visual perception. The no-tool condition isolates internal solution ability but does not estimate performance of web-enabled or code-enabled agents. Neither condition is intrinsically superior; they answer different questions.

Judge dependence. Blind single-answer Codex judging is scalable and auditable but not independent of the OpenAI ecosystem. Every flagged substantive doubt received human adjudication, but this authority mechanism is not an independent judge–human agreement study. Frozen criteria, evidence, validation, and human review reduce rather than eliminate judge bias; a stratified agreement study and a diverse judge panel are important next steps.

Operational resources. Resource figures support comparisons within this campaign, not universal cost or compute claims. Token fields differ across providers, and Fable combines visible output with hidden thinking. The \$84.21 value is the equal-weight, replicate-normalized panel estimate, whereas \$105.10 sums all retained successful responses before within-cell averaging; neither is a provider bill, because failed attempts and out-of-panel calls are not fully observed. Prices are dated, Gemma’s zero is campaign-specific, and latency includes network, provider, backoff, and retry effects from one execution period.

Date-based temporal evidence. Dating an item after a provider-reported cutoff reduces one exposure pathway but cannot rule out all forms of contamination. Proprietary endpoints and backend routing can also change after the snapshot. Reproduction therefore means reconstructing the preserved campaign, not assuming that a later endpoint will regenerate identical answers.

8 Conclusion

This benchmark links authentic university exercises to a complete evidence chain: direct first-party no-tool generation, separate raw and normalized artifacts, frozen atomic rubrics, blind single-answer judgments, deterministic validation, versioned human authority resolution, and exact unblinding. Its balanced 14-by-60 analysis contains 840 paired cells derived from 924 retained successful responses.

The empirical conclusion is more informative than a single rank. Six configurations lie between 96.71% and 98.72%, with paired uncertainty that does not support a sharply separated leader. Yet the same systems differ substantially in domain stability, token volume, price, and response time. The same-weights DeepSeek contrast further shows that a large reasoning gain can carry a substantial resource penalty. These results make correctness a threshold rather than a sufficient selection rule. For near-ceiling scientific problem solving, the central outcomes are completeness, cross-domain robustness, and observable quality–resource trade-offs.

Data and Reproducibility

The private archive preserves sources, solutions, approved rubrics, prompts, configurations, raw and normalized responses, metadata, blind maps, judgments, adjudication history, and authority-resolved indexes. This repository releases sanitized panels, tables, figure generators, methodology, schemas, templates, and manuscript sources. Only valid, cur-

Table 6. Provider-reported temporal metadata and conservative date-based item counts. “Later items” compares the dated examination item with the stated cutoff, not with model release. Knowledge and training labels remain distinct.

Configuration(s)	Provider label	Reported date	Later items	Official source
GPT-4.1; o3	Knowledge cutoff	1 Jun 2024	60 / 60	OpenAI model documentation (OpenAI, 2026c,b)
Gemini 3.1 Pro; Gemini 3.5 Flash	Knowledge cutoff	Jan 2025	54 / 60	Google model documentation (Google, 2026a,b)
Claude Sonnet 4.5	Reliable knowledge cutoff	Jan 2025	54 / 60	Anthropic models overview (Anthropic, 2026a)
Claude Sonnet 4.5	Public-Internet component date	Jul 2025	36 / 60	Anthropic Transparency Hub (Anthropic, 2026b)
Claude Haiku 4.5	Reliable knowledge / training data	Feb 2025 / Jul 2025	48 / 60; 36 / 60	Anthropic overview and Transparency Hub (Anthropic, 2026a,b)
Gemma 4 31B IT	Pre-training dataset cutoff	Jan 2025	54 / 60	Gemma 4 model card (Google, 2026c)
Claude Fable 5	Reliable knowledge / training data	Jan 2026	21 / 60	Anthropic models overview (Anthropic, 2026a)
Grok 4.5	Knowledge cutoff	1 Feb 2026	21 / 60	xAI model documentation (xAI, 2026)
GPT-5.6 Sol	Knowledge cutoff	16 Feb 2026	12 / 60	OpenAI model documentation (OpenAI, 2026a)
DeepSeek V4; Grok 4.3	No model-specific cutoff located	–	–	No post-cutoff count assigned

rent, unblinded judgments with matching response hashes enter results; upstream regeneration requires the private archive. Credentials are excluded.

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